

BIRLA INSTITUTE OF TECHNOLOGY MESRA, RANCHI
DEPARTMENT OF MATHEMATICS

MA24103 Mathematics-II

Session: SP 2026

Tutorial - 4 (Module IV)

1. Check the analyticity of the following functions:

- a) $f(z) = z^2 + 2z + 3$
- b) $f(z) = |z|^2$
- c) $f(z) = e^z$
- d) $f(z) = x^2 + iy^2$
- e) $f(z) = e^{-x}(\cos y - i \sin y)$

2. Show that each of these functions is nowhere analytic

- a) $f(z) = z + \bar{z}$
- b) $f(z) = |z|^2$
- c) $f(z) = y + ix$
- d) $f(z) = 2xy + i(x^2 - y^2)$
- e) $f(z) = e^{y+ix}$

3. Prove that the function $f(z) = \begin{cases} \frac{z^5}{|z|^4}, & z \neq 0 \\ 0, & z = 0 \end{cases}$ satisfies the C-R equations at $z = 0$, but is not differentiable there.

4. Let $u(x, y)$ and $v(x, y)$ denote the real and imaginary components of the function $f(z)$ defined by

$$f(z) = \begin{cases} \frac{\bar{z}^2}{z} & \text{when } z \neq 0 \\ 0 & \text{when } z = 0 \end{cases}$$

Verify that the Cauchy-Riemann equations are satisfied at the origin $z = (0, 0)$.

5. Show that $u(x, y)$ is harmonic in some domain and find a harmonic conjugate $v(x, y)$ when

- a) $u(x, y) = x^2 - y^2 - y$
- b) $u(x, y) = x^3 - 3x^2y - 3xy^2 + y^3$

6. Apply Cauchy's theorem to show that $\int_C f(z)dz = 0$ when the contour C is the unit circle $|z| = 1$, in either direction, and when

- a) $f(z) = \frac{z^2}{z-3}$
- b) $f(z) = ze^{-z}$
- c) $f(z) = \frac{1}{z^2 + z + 2}$

7. Find the value of the integral

- a) $\int_{|z|=1} \frac{z \sin z}{z - \pi} dz$
- b) $\int_{|z|=1} \frac{1}{z+2} dz$
- c) $\int_{|z|=1} \frac{e^z}{z^3} dz$
- d) $\int_{|z|=2} \frac{1}{z^2(z-3)} dz$
- e) $\int_{|z|=2} \frac{\sin z}{(z^2+1)} dz$

8. Find the radius of convergence of the following power series

- a) $\sum_{n=0}^{\infty} (1 + \frac{1}{n})^{n^2} z^n$
- b) $\sum_{n=0}^{\infty} n! z^n$
- c) $\sum_{n=0}^{\infty} \frac{z^n}{n!}$
- d) $\sum_{n=0}^{\infty} \frac{n^n}{n!} z^n$

9. Find the residue at $z = 0$ of the function

- a) $\frac{1}{z^2}$
- b) $\frac{1}{z + z^2}$
- c) $\frac{z - \sin z}{z^3}$
- d) $\frac{e^z}{z^3}$
- e) $e^{\frac{1}{z}}$

10. Use Cauchy's residue theorem to evaluate the integrals of each of these functions around the circle $|z| = 3$ in the positive sense

a) $\frac{e^{-z}}{z^2}$

b) $\frac{z+1}{z(z-2)}$

c) $z^2 e^{\frac{1}{z}}$

Solution:

5. a) $v(x, y) = 2xy + x$

b) $v(x, y) = x^3 + 3x^2y - 3xy^2 - y^3 + C$

7. a) 0

b) 0

c) πi

d) $\frac{-2\pi i}{9}$

e) 0

8. a) e

b) 0

c) ∞

d) e

9. a) 0

b) 1

c) $\frac{1}{6}$

d) $\frac{1}{2}$

e) 1

10. a) $-\pi i$

b) $-2\pi i$

c) $\frac{\pi i}{3}$